

Unit 6: Relations and Graphs

Relations

The word relation is used to indicate a relationship between two objects. Relationship between elements of sets is represented using a structure called a relation. Relations can be used to solve problems such as determining which pair of the cities are linked by airline flights in a network, producing a useful way to store information in computer database, to show the relationship between employee and his or her salary in database etc.

In mathematics, an example of relation is 'less than' which is denoted by $<$, so that x is related to y if $x < y$, two computer programs are related if they share the same common data.

Definition

Let A and B be two non-empty sets. A relation from A to B is any subset of the Cartesian product $A \times B$ satisfying given specific condition.

$$R \subseteq A \times B$$

Suppose that R is a relation from A to B . Then R is a set of ordered pairs (a,b) where $a \in A$ and $b \in B$. Every ordered pairs (a,b) is written as $a R b$, read as 'a is related to b by R'. If $(a,b) \notin R$, then a is not related to b by R and written as $a \not R b$. If R is a relation from a set T to itself, that is, if R is subset of $A^2 = A \times A$ then we say that R is relation on A .

Domain and Range

Suppose that R is a relation from A to B then, the domain of relation R is the set of all first elements of ordered pairs which belongs to R . It is denoted by $\text{Dom}(R)$.

Mathematically, $\text{Dom}(R) = \{a \in A : (a,b) \in R, \text{ for some } b \in B\}$.

Similarly, the range of relation R is the set of all second elements of ordered pairs belongs to relation R .

Mathematically, $\text{range}(R) = \{b \in B : (a,b) \in R, \text{ for some } a \in A\}$

If $R = \{(1,2), (1,4), (2,4)\}$ then

$$\text{Dom}(R) = \{1,2\}$$

$$\text{Ran}(R) = \{2,4\}$$

Example: Let $A = \{4,5,6\}$, find the relations in $A \times A$ under the condition $x+y < 10$. Also find the domain and range of relation.

Solution: $A \times A = \{(4,4), (4,5), (4,6), (5,4), (5,5), (5,6), (6,4), (6,5), (6,6)\}$

The given condition is $x+y < 10$

So, $R = \{(4,4), (5,4), (5,4)\}$

$\text{Dom}(R) = \{4,5\}$

$\text{Range}(R) = \{4,5\}$

Properties of Relation

A relation R on a set A satisfies certain properties which are defined as follows:

Reflexive Relation: A relation R on a set A is reflexive if $(a,a) \in R$ for all $a \in A$, that is, if aRa for all $a \in A$. A relation R on a set A is irreflexive if aRa for every $a \in A$

Example : If $R = \{(1,1), (1,2), (2,2), (2,3), (3,3)\}$ is a relation on $A = \{1,2,3\}$ then R is a reflexive relation since for every $a \in A$, $(a,a) \in R$

Symmetric Relation: A relation R on a set A is symmetric if $(b,a) \in R$ whenever $(a,b) \in R$ for all $a, b \in A$.

Asymmetric Relation

A relation R on a set A is asymmetric if $(a,b) \in R$ then $(b,a) \notin R$ for all $a, b \in A$.

Antisymmetric Relation: A relation R on a set A is antisymmetric if $a = b$ whenever aRb and bRa

Transitive Relation: A relation R on a set A is transitive if whenever aRb and bRc then aRc . That is, $(a,b) \in R$ and $(b,c) \in R \rightarrow (a,c) \in R$ for all $a, b, c \in R$

Example : Let $A = \{1,2,3\}$ and $R = \{(1,2), (3,2), (2,3), (1,3), (2,2), (3,3)\}$ then R is transitive.

Combining Relations: The relation from set A to B is a subset of $A \times B$. So two relations from A to B can be combined in a same way that two sets can be combined.

Example: Let $A = \{4,5,6\}$ and $B = \{4,5,6,7\}$. The relation $R_1 = \{(4,4), (5,5), (6,6)\}$ and $R_2 = \{(4,4), (4,5), (4,6), (4,7)\}$ and then

Find

- $R_1 \cup R_2$
- $R_1 \cap R_2$
- $R_1 - R_2$
- $R_2 - R_1$

Solution:

- $R_1 \cup R_2 = \{(4,4), (5,5), (6,6)\} \cup \{(4,4), (4,5), (4,6), (4,7)\} = \{(4,4), (5,5), (6,6), (4,5), (4,6), (4,7)\}$
- $R_1 \cap R_2 = \{(4,4), (5,5), (6,6)\} \cap \{(4,4), (4,5), (4,6), (4,7)\} = \{(4,4)\}$
- $R_1 - R_2 = \{(4,4), (5,5), (6,6)\} - \{(4,4), (4,5), (4,6), (4,7)\} = \{(5,5), (6,6)\}$
- $R_2 - R_1 = \{(4,5), (4,6), (4,7)\}$

Types of Relations

Complementary Relation: Let R be a relation from set A to B . The complementary relation of R is denoted by R^c which consists of those ordered pairs which are not in R .

That is

$$R^c = \{(a,b) \in A \times B : (a,b) \notin R\}$$

Example: Let R be a relation on set $A = \{1,2,6\}$ defined as $R = \{(x,y):x < y\}$. Find the complement relation of R .

$$\text{Solution: } A \times A = \{(1,1),(1,2),(1,6),(2,1),(2,2),(6,1),(6,2),(6,6)\}$$

$$R = \{(1,2),(1,6),(2,6)\}$$

$$R^c = A \times A - R = \{(1,1),(2,1),(2,2),(6,1),(6,2),(6,6)\}$$

Inverse Relation : Let R be a relation from A to B , the inverse of R , denoted by R^{-1} , is the relation from B to A which consists of those ordered pairs which, when reversed, belong to R ;

That is,

$$R^{-1} = \{(b,a) : (a,b) \in R\}$$

Example: Let $A = \{1,2,3,4\}$ and $B = \{a,b,c\}$

And Let $R = \{(1,a),(1,b),(2,b),(2,c),(3,b),(4,a)\}$

Compute R^c and R^{-1}

Solution:

$$A \times B = \{(1,a),(1,b),(1,c),(2,a),(2,b),(2,c),(3,a),(3,b),(3,c),(4,a),(4,b),(4,c)\}$$

$$R^c = \{(1,c),(2,a),(3,a),(3,c),(4,b),(4,b),(4,c)\}$$

$$R^{-1} = \{(a,1),(b,1),(b,2),(c,2),(b,3),(a,4)\}$$

Identity Relation

A relation R in a set A i.e a relation A to A is said to be a identity relation, generally denoted by I_A if,

$$I_A = \{(x,x):x \in A\}$$

Example: Let $A = \{1,2,3\}$ then $I_A = \{(1,1),(2,2),(3,3)\}$ is an identity relation

N-ary Relation: Relationship among elements of more than two sets often arises. The relationship among more than two sets is called n-ary relations.

For instance, there is a relationship involving the airline, flight number, starting point, destination, departure time, arrival time of a flight. N-ary relations are used to represent computer databases which help us answer queries about the information stored in database such as which flights land at Kathmandu airport between 1 pm to 3 pm?

Let $A_1, A_2, A_3, \dots, A_n$ be a finite sets. A subset R of $A_1 \times A_2 \times A_3 \times \dots \times A_n$ is called an n-ary relation on $A_1, A_2, \dots, A_n$.

Example

Let $A = \{1, 2\}$ and let R be the relation defined by the property ' $x + y + z$ ' is even.

Now $A \times A \times A = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2), (1, 2, 2)\}$

Under ' $x + y + z$ is even' the relation is

$R = \{(1, 1, 2), (1, 2, 1), (2, 1, 1), (2, 2, 2)\}$ is ternary relation

Example

Let R be the relation consisting of 5-tuples (A, N, S, D, T) representing airplane flights, where A is the airline, N is the flight number, S is the starting point, D is the destination point, and T is the departure time. The ordered pairs belongs to such n-ary relation may be: (Fly Dubai, 1024, Kathmandu, Dubai, 2:00)

Operations on N-ary Relation

There are variety of operations on N-ary relations that can be used to form new nary relations. The most common operation is determining all N-tuples in N-ary relation that satisfy certain conditions

1. Let R be an n-ary relation and C a condition that elements in R may satisfy. Then the selection operator S_c maps the N-ary relation R to the N-ary relation of an n-tuples from R that satisfy the condition C
2. The projection $\Pi_1, \Pi_2, \Pi_3, \dots, \Pi_m$ maps the N-tuple $(a_1, a_2, a_3, \dots, a_n)$ to the m-tuple $(a_{i_1}, a_{i_2}, a_{i_3}, \dots, a_{i_m})$ where $m \leq n$
3. Let R be a relation of degree m and S a relation of degree n . The join $J_p(R, S)$ where $p \leq m$ and $p \leq n$ is a relation of degree $m+n-p$ that consists of all $(m+n-p)$ tuples $(a_1, a_2, \dots, a_{m-p}, c_1, c_2, \dots, c_p, b_1, b_2, \dots, b_{n-p})$ where the tuple $(a_1, a_2, \dots, a_{m-p}, c_1, c_2, \dots, c_p)$ belongs to R and the n-tuple $(c_1, c_2, c_3, \dots, c_{n-p})$ belongs to S

Example:

Suppose that we have following class schedule for the BSc course in a particular college

Department	Course-number	Room	Time
Computer Science	415	C101	11:00
Mathematics	416	C102	12:00

Mathematics	417	C104	13:00
Physics	418	C104	14:00
Statistics	419	C105	10:00
Computer Science	420	C103	13:00

Now we can use selection operation in this N-ary relation(N=4) as follows:

SELECT room, time FROM class-schedule WHERE department = "Computer Science"

The output of above selection operation results the following table

Room	Time
C101	11:00
C103	13:00

Composition of Relation

Let A, B, C be three sets. Let R be a relation from A to B and S be a relation from B to C. Then the composite of R and S is denoted by SoR and defined as

$SoR = \{(a, c) \in A \times C \text{ for some } b \in B, (a, b) \in R \text{ and } (b, c) \in S\}$

That is a (SoR) c, if form some $b \in B$, we have aRb and bSc

Example : If R and S be relation on $A = \{1, 2, 3, 4\}$ defined by $R = \{(1, 1), (1, 2), (3, 4), (4, 2)\}$ and $S = \{(1, 1), (2, 1), (3, 1), (4, 4), (2, 2)\}$. Find RoS.

Solution:

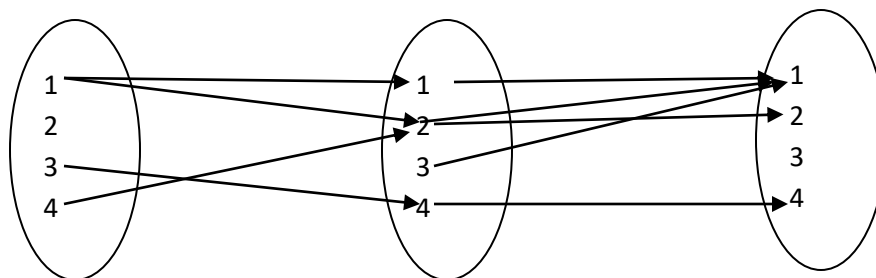
Given, $A = \{1, 2, 3\}$

$R: A \rightarrow A = \{(1, 1), (1, 2), (3, 4), (4, 2)\}$

$S: A \rightarrow A = \{(1, 1), (2, 1), (3, 1), (4, 4), (2, 2)\}$

Now,

For SoR,



From above diagram,

1. $1 \rightarrow 1 \rightarrow 1 : (1,1) \in \text{SoR}$
2. $1 \rightarrow 2 \rightarrow 1 : (1,1) \in \text{SoR}$
3. $1 \rightarrow 2 \rightarrow 2 : (1,2) \in \text{SoR}$
4. $4 \rightarrow 1 \rightarrow 1 : (4,1) \in \text{SoR}$
5. $4 \rightarrow 2 \rightarrow 2 : (4,2) \in \text{SoR}$
6. $3 \rightarrow 4 \rightarrow 4 : (3,4) \in \text{SoR}$

Example: Let $A = \{1, 2, 3\}$, $B = \{p, q, r\}$, $C = \{x, y, z\}$ and Let $R = \{(1,p), (1,r), (2,q), (3,q)\}$ and $S = \{(p,y), (q,x), (r,z)\}$. Compute SoR

Solution:

Representations of Relations

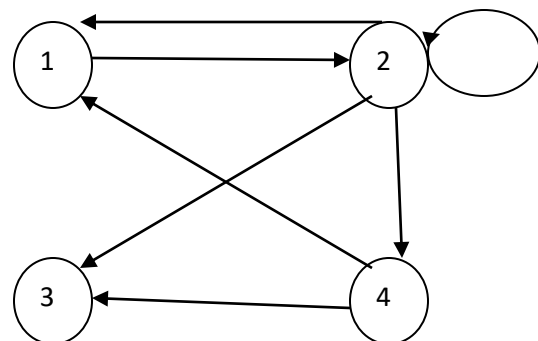
There are many ways of representing relations on finite sets. For visualizing the information about relations, graphical methods are particularly useful

Directed Graphs of Relations

Let A and B be two finite sets and R be a relation from A to B . For graphical representation of relation on a set, each element of the set is represented by a point. These points are called nodes or vertices. An arc is drawn from one point to its related point. If the pair $x \in A, y \in B$ is in the relation, the corresponding nodes are connected by arcs called edges or arcs. The arcs start at the first element of the pair and they go to the second element of the pair. This direction is indicated by an arrow. All the arcs with an arrow are called directed arcs. The resulting pictorial representation of R is called a directed graph or digraph of R . An edge of the form (a,a) is represented using an arc from the vertex 'a' back to itself. Such an edge is called a loop.

Example: Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (2,4), (3,4), (4,1)\}$

Then the diagram of R is shown as in figure below:



Matrix of a Relation

Let $A = \{a_1, a_2, a_3, \dots, a_n\}$ and $B = \{b_1, b_2, \dots, b_n\}$ be finite sets containing m and n elements respectively and let R be a relation from A to B then R can be represented by mn matrix $MR = [m_{ij}]_{mn}$ where

$m_{ij} = 1$ if $(a_i, b_j) \in R$

0 otherwise

Then the matrix MR is called matrix of relation R

Reflexive, Symmetric and Transitive Closure of Relations

Reflexive Closure of R

Let R is a relation then any new relation R' that contains R and is also contained within every reflexive relation that contains R is called the reflexive closure of R

Given a relation R on a set A , the reflexive closure of R can be formed by adding to R all pairs of the form (a, a) with $a \in A$, which is already in R

Example: Given $R = \{(1,1), (1,2), (2,1), (3,2)\}$ on the set $A = \{1,2\}$ is not reflexive.

Now this relation can be made reflexive by adding $(2,2)$ and $(3,3)$ to R . Since these are the only pairs of the form (a,a) not in R .

Then

$R' = \{(1, 1), (1, 2), (2, 1), (3, 2), (2, 2), (3, 3)\}$

Symmetric Closure of R

The symmetric closure of relation R on set A is the smallest symmetric relation. On set A that contains given relation R .

Example: For finite set, we can construct transitive closure of R step by step, starting from R and adding transitive edges

That is, $R^+ = \bigcup_{i \in \{1, 2, 3, \dots\}} R^i$

Where R_i is the i^{th} power of R , defined inductively as

$R^1 = R$

And $R^{i+1} = R \circ R^i$ for $i > 0$

Where ' $\circ$ ' denotes the composition of relations

Equivalence Relation

A relation **R** on a set **A** is called an equivalence relation if it is reflexive, symmetrical and transitive

Example: Let $A = \{1, 2, 3, 4, 5\}$. Show that the relation $R = \{(1,1), (1,5), (2,2), (2,4), (3,3), (4,2), (4,4), (5,1), (5,5)\}$ is a equivalence relation

Solution:

Given $A = \{1, 2, 3, 4, 5\}$

$R = \{(1,1), (1,5), (2,2), (2,4), (3,3), (4,2), (4,4), (5,1), (5,5)\}$

To prove R is equivalence, we need to show that R satisfies reflexive, symmetric and transitive property.

- Reflexive: $\forall a \in A (a, a) \in R$. Here in relation R, for all $1, 2, 3, 4, 5 \in A$, we have $(1,1) \in R, (2,2) \in R, (3,3) \in R, (4,4) \in R$ and $(5,5) \in R$
- Symmetric: $1, 5 \in A. (1, 5) \in R \rightarrow (5,1) \in R. 2, 4 \in A, (2,4) \in R \rightarrow (4,2) \in R$. R is symmetric
- Transitive: $\forall a, b, c \in A, (a,b) \in R$ and $(b,c) \in R \rightarrow (a, c) \in R$. In given relation R, $1, 2, 3, 4, 5 \in A$,
 $(1,5) \in R \wedge (5,1) \in R \rightarrow (1,1) \in R$
 $(2,4) \in R \wedge (4,2) \in R \rightarrow (2,2) \in R$
 $(2,4) \in R \wedge (4,4) \in R \rightarrow (2, 4) \in R$
 $(5,1) \in R \wedge (1,5) \in R \rightarrow (5,5) \in R$

Congruence Modulo Relation

Let a and b be two integers then a is congruence to b modulo m is m divides a-b. We use notation $a \equiv b \pmod{m}$

For example if $a=20, b=10$ and $m=5$ then $20 \equiv 10 \pmod{5}$

Example 1: Let m be a positive integer with $m > 1$ show that the relation $R = \{(a, b): a \equiv b \pmod{m}\}$ is an equivalence relation on set of positive integers

Solution: To prove that R is equivalence, we have to show that R is reflexive, symmetric and transitive properties.

- Reflexive:** $\forall a \in \mathbb{Z}^+, a \equiv a \pmod{m} \rightarrow m|(a-a) \rightarrow m|0$ (True). Hence $(a, a) \in R$. So R is reflexive
- Symmetric :** $\forall a, b \in \mathbb{Z}^+$, let $(a,b) \in R$
So $a \equiv b \pmod{m} \rightarrow m|(a-b)$
 $a-b = mk$ or $b-a = m(-k)$
 $b \equiv a \pmod{m}$
Therefore $(b,a) \in R$

iii. **Transitive:** $\forall a, b, c \in \mathbb{Z}^+$. Let $(a, b) \in R$.

$$\text{So } a \equiv b \pmod{m} \rightarrow m|(a-b)$$

$$a-b = mk_1 \dots\dots\dots(1)$$

Again let $(b, c) \in R$

$$\text{So } b \equiv c \pmod{m} \rightarrow m|(b-c)$$

$$b-c = mk_2 \dots\dots\dots(2)$$

Adding (1) and (2) we get

$$a-b + b-c = mk_1 + mk_2$$

$$\text{or } a-c = m(k_1 + k_2)$$

$$\text{or } a-c = mk$$

$$\rightarrow m|(a-c)$$

$$\rightarrow a \equiv c \pmod{m}$$

$$(a, c) \in R$$

Therefore R is transitive

Example 2: Show that the relation $R = \{(x, y) : x \equiv y \pmod{4}\}$ defined on the set $A = \{1, 2, 3, 4, \dots, 13, 14, 15\}$ is an equivalence. Also find the partition of the set A determined by relation R

Solution:

To prove that R is equivalence, we have to show that R is reflexive, symmetric and transitive properties.

i. **Reflexive:** $\forall x \in A, x \equiv x \pmod{m} \rightarrow 4|(x-x) \rightarrow 4|0$ (True). Hence $(x, x) \in R$. So R is reflexive

ii. **Symmetric :** $\forall x, y \in A$, let $(x, y) \in R$

$$\text{So } x \equiv y \pmod{4} \rightarrow 4|(x-y)$$

$$x-y = 4k \text{ or } y-x = 4(-k)$$

$$y \equiv x \pmod{4}$$

$$\text{Therefore } (y, x) \in R$$

iii. **Transitive:** $\forall x, y, z \in A$. Let $(x, y) \in R$.

$$\text{So } x \equiv y \pmod{4} \rightarrow 4|(x-y)$$

$$x-y = m.4 \dots\dots\dots(1)$$

Again let $(y, z) \in A$

$$\text{So } y \equiv z \pmod{4} \rightarrow 4|(y-z)$$

$$y-z = 4m \dots\dots\dots(2)$$

Adding (1) and (2) we get

$$x-y + y-z = 4.m + 4.m$$

$$\text{or } x-z = m(4+4)$$

$$\begin{aligned}
 &\text{or } x-z = 8m \\
 &\rightarrow 4 \mid (x-z) \\
 &\rightarrow x \equiv z \pmod{4} \\
 &(x, z) \in R
 \end{aligned}$$

Therefore R is transitive

Now, to find the partition of set A determined by relation **R**, we need to find the modulo classes. The possible remainders after performing modulo 4 are 0,1,2,3 so.

$$[0]_4 = \{4, 8, 12\}$$

$$[1]_4 = \{1, 5, 9, 13\}$$

$$[2]_4 = \{2, 6, 10, 14\}$$

$$[3]_4 = \{3, 7, 11, 15\}$$

$$\text{Therefore } P = \{\{4,8,12\},\{1,5,9,13\},\{2,6,10,14\},\{3,7,11,15\}\}$$

Example 3: What are the equivalence classes of 0 and 1 for congruence modulo 4?

Solution: The equivalence class of an integer a modulo m is defined by $[a]_m$ so that

$$[a]_m = \{\dots a-2m, a-m, a, a+m, a+2m, \dots\}$$

Now, the equivalence class of 0 contains all integers a such that $a \equiv 0 \pmod{4}$. The integers in this case are those divisible by 4

$$[0]_4 = \{\dots -8, -4, 0, 4, 8, \dots\}$$

The equivalence class of 1 contains all integers a such that $a \equiv 1 \pmod{4}$. That is, those that have remainder 1 when divided by 4

$$\text{Therefore } [1]_4 = \{\dots -7, -3, 1, 5, 9, \dots\}$$

Example 4: What are the sets in the partition of integers arising from congruence modulo 4?

Solution: Since the congruence classes are disjoint so, congruence classes form a partition. The congruence class modulo 4 are denoted by $[0]_4$, $[1]_4$, $[2]_4$ and $[3]_4$

$$\text{Therefore } [0]_4 = \{\dots -8, -4, 0, 4, 8, \dots\}$$

$$[1]_4 = \{\dots -7, -3, 1, 5, 9, \dots\}$$

$$[2]_4 = \{\dots -6, -2, 2, 6, 10, \dots\}$$

$$[3]_4 = \{-5, -1, 3, 7, 11, \dots\}$$

Equivalence classes:

If R is an equivalence relation on a set A and xRy then x and y are called equivalent with respect to R . Then the class of any element $[x]_R = \{y \in A : (x, y) \in R\}$

The collection of all equivalence classes of elements under an equivalence relation R is denoted by A/R that is,

$$A/R = \{[x] : x \in A\}.$$

It is called quotient set of A by R

Example 1: Let $R = \{(1,2), (2,1), (1,1), (2,2), (3,3), (4,4)\}$ be a relation on $A = \{1,2,3,4\}$. Find the equivalence classes of each element of A and quotient set of A by R .

Solution:

Equivalence class of 1 is:

$$[1]_R = \{1,2\}$$

$$[2]_R = \{1,2\}$$

$$[3]_R = \{3\}$$

$$[4]_R = \{4\}$$

$$\text{And } A/R = \{\{1,2\}, \{3\}, \{4\}\}$$

From..

Partition of a Set

A partition or quotient set of a set A is the collection of subsets of A . That is

$$P = \{A_1, A_2, \dots, A_n\} \text{ such that}$$

- i. Union of A_i is A
- ii. For distinct A_i and A_j , $A_i \cap A_j = \phi$

The sets in P are called blocks or cells of partition

Example: Let $A = \{a, b, c, d, e, f, g, h\}$ and $P = \{A_1, A_2, A_3, A_4, A_5\}$ where

$$A_1 = \{a, b, c, d\}$$

$$A_2 = \{a, c, e, f, g, h\}$$

$$A_3 = \{a, c, e, g\}$$

$$A_4 = \{b, d\}$$

$$A_5 = \{f, h\}$$

Then $\{A_1, A_2\}$ is not a partition since $A_1 \cap A_2 = \phi$

Also $\{A_1, A_5\}$ is not a partition

But $P = \{A_3, A_4, A_5\}$ is a partition of A since $A_3 \cup A_4 \cup A_5 = A$ and $A_3 \cap A_4 = \phi$, $A_4 \cap A_5 = \phi$ and $A_3 \cap A_5 = \phi$

Graph Theory

Introduction

Many situations that occur in computer science, physical science, communication science, Economics and many others areas can be analyzed by using techniques found in a relatively new area of mathematics called graph theory. Graphs can be used to represent almost any problem involving discrete arrangements of objects where concern lies not with the internal properties of these objects but with relationships among them

Graph is a discrete structure consisting of vertices and edges connecting the vertices. A graph $G = (V, E)$ is a mathematical model consists of two non empty sets: $V = \{v_1, v_2, v_3, \dots, v_n\}$ is called set of vertices and $E = \{e_1, e_2, e_3, \dots, e_n\}$ of ordered or unordered pairs of distinct vertices called edges. We usually write $G = (V, E)$ and say V is the vertex and E is the edge set of G

Applications of Graph

Graphs are used to solve problems in many fields. Some of them are as follows:

- Graphs are used to model the geographic maps of cities in which each place in city can be represented by the node and the road connecting such places are represents the edge
- Graphs are used to model the computer network in which each node is a machine and link between them represent the edge
- They are used to analyze the electric circuits, project planning, genetics etc
- Any structured problem can be modeled by graphs. Then can help to solve typical problems those connected with finding shortest path of most economical route between two vertices, or the smallest set of edges which connect all the vertices in a graph
- Graphs are used to study the structure of WWW
- It is used to find the number of different combination of flight between two cities in an airline network
- It can be used to distinguish between two chemical compounds with the same molecular formula but different structure
- It is also used to assign channels to television stations

Simple Graph

We define a simple graph as 2-tuple consisting of a non- empty set of vertices V and a set of unordered pairs of distinct elements of vertices called edges, having neither loop not parallel edges. A graph $G = (V,E)$ is said to be simple if G has no loops and no parallel edges.

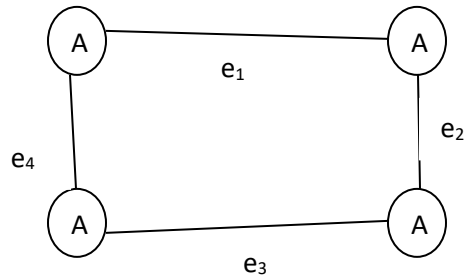


Figure Simple Graph

Multi-Graph: A computer network may contain multiple links between data centers, to model such network we need graphs that have more than one edge connecting the same pair of vertices. Two or more edges between same pair of vertices are called parallel edges or two or more edges having same end points are called parallel edges

Pseudograph

Sometimes a communication link connects a data center with itself, perhaps a feedback loop for diagnostic purpose, to model such a network we need to include edges that connects a vertex to itself such an edge is called a loop

Graph Models

Graph Terminologies

- **Order and Size of Graph:** If $G = (V,E)$ be a finite graph then the number of vertices in graph G is called order of graph G and the number of edges in G is called size of G
- **Degree of Vertex:** Let $G = (V,E)$ be a graph and v be a vertex of G . The degree of v is denoted by $d(v)$ is the number of edges incident on v
- **Isolated and pendant Vertex:** In a graph $G = (V,E)$, a vertex having degree zero (0) is called isolated vertex and vertex having degree one (1) is called pendant vertex

Degree sequence of a Graph: If $v_1, v_2, v_3, \dots, v_n$ are n vertices of G then the sequence $(d_1, d_2, d_3, \dots, d_n)$ where $d_i = \deg(v_i)$ is the degree sequence of G . In general, we order the vertices so that the degree sequence is monotonically increasing.

That is

$$d(G) = d_1 \leq d_2 \leq d_3 \dots \leq d_n = \Delta(G)$$

where $d(G) = \min \{ \deg v : v \in V(G) \}$

and $\Delta(G) = \max \{ \deg v : v \in V(G) \}$

Adjacent vertices and adjacent edges

Two vertices u and v in an unordered graph are called adjacent if there is an edge between u and v .

Two edges e_1 and e_2 in an undirected graph are said to be adjacent if they share a common vertex.

Simple and Special Graph

Some important types of graphs are introduced here. These graphs are often used as examples and arise in many applications.

Trivial Graph: A graph with one vertex and no edges is called a trivial graph

Complete Graph: A graph is said to be complete if there exists exactly one edge between any pair of vertices in G . The complete graph with n vertices is denoted by K_n .

Round Robin Tournament: A game tournament where each team plays with each other exactly once is called round-robin tournament. Such tournaments can be modeled using complete graph where there exists exactly one edge between each possible pair of vertices except itself. The following figure shows the tournament between 5 teams in Round-Robin fashion. Here, in the graph edge between vertices represent the matches between the teams and vertices represent teams.

Collaborative Graph

A simple graph that can be used to model joint authorship of academic papers, where vertices represent people and edges between two vertices represent joint authorship if they have jointly written paper.

Call Graph: Graph can be used to model telephone calls in a telecommunication network. In such graph, each telephone number is represented by a vertex and each telephone call is represented by edge. The source vertex represents a start call (sender) and the vertex where edge ends, represents the end of call (receiver)

Regular Graph: A graph in which all vertices are of same (equal) degree is called a regular graph. If the degree of each vertex is r then the graph is a r -regular graph.

Bipartite Graph: A graph $G = (V, E)$ is said to be bipartite if V is the union of two non-empty disjoint subsets V_1 and V_2 of V such that each edge in E is incident on one vertex in V_1 and one vertex in V_2

Or

A graph $G = (V, E)$ is said to be bipartite if its vertex set V can be partitioned into two subsets V_1 and V_2 such that each edge of G connects the vertex of V_1 to vertex of V_2 so that no edge in G connects two vertices in V_1 or two vertices in V_2 .

Complete Bipartite Graph : A bipartite graph is said to be complete if each vertex of first bipartite set is connected to every vertex of another bipartite set. A complete bipartite graph with m number of vertices in first partite set and n number of vertices in second bipartite set is denoted $K_{m,n}$

Cycle Graph: A graph $G = (V, E)$ with vertices $n \geq 3$ in which every vertex is connected with two other vertices one on either side of it and last vertex is connected with first one to form a closed path, is called cycle graph

Binary Tree

A binary tree is a finite set of elements that is either empty or partitioned into three disjoint subsets. The first subset contains the single element called root of the tree, the other subsets are called left and right sub tree of original tree. The binary tree is so named because each node can have at most two descendants.

Types of binary trees

Strictly Binary Tree

It is a binary tree with non-empty right and left sub trees. In other words, it is binary tree with every node N has either 0 or 2 children. The strictly binary tree is also known as extended 2 Tree or simply 2 –tree. Sometimes nodes with 2 children's are known as internal nodes and nodes with 0 children are known as external nodes.

Complete Binary Tree

It is a special type of strictly binary tree where all the leaves of the tree reside at the same level. Using the depth we always say a complete binary tree of depth d where all leaves will be at level d

Binary Search Tree

The application of binary tree is searching and sorting. By enforcing certain rules on the values of the elements stored in a binary tree, it could be used to search and sort.

Binary search tree is a tree in which value of each node in the tree is greater than the value of node in its left (if exists) and it is less than the value in its right child (if it exists).

As name suggests, binary search tree (BST) is used for searching purpose. For every node N in the BST, the following property will be true.

- The data at left node will be smaller than data at node N
- The data at right node will be larger than data at node N

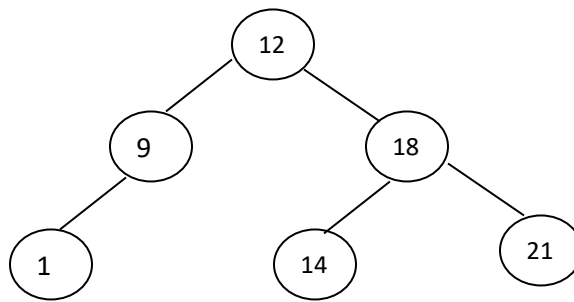
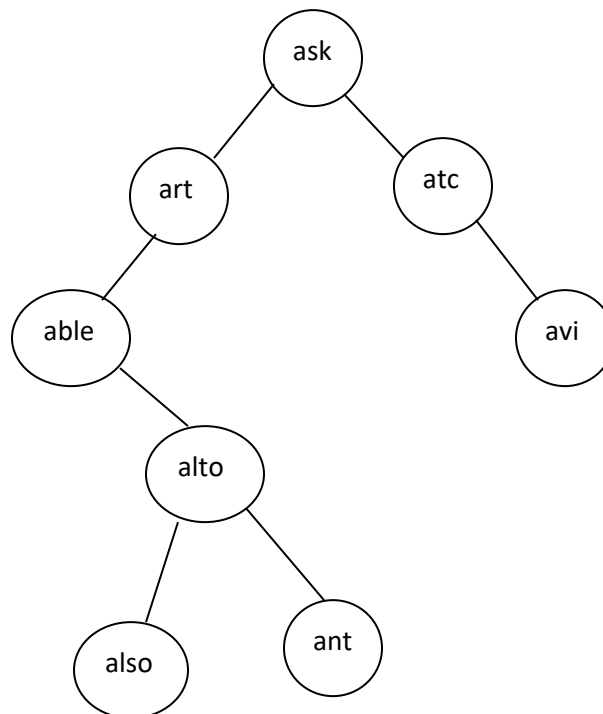


Figure: Binary Search Tree

Example: Draw a BST for the following words: ask, art, atc, able, alto, also, avoid, ant

Solution

The first key in the list act as root key for node in the tree



Expression Tree

Binary trees are used to represent arithmetic expressions. An expression tree is a binary tree used to represent mathematical expression in which all the leaf node contains operands such as constants or variables and non leaf node contains mathematical operators.

Example: Use binary expression tree to represent the expressions

- a. $a-b$
- b. $(a+b)*c$
- c. $(a+b)*(c-d)$
- d. $((a+b)*c)+(d/e)$
- e. $(a+b)*(c+(d/e))$

M-ary Tree

A rooted tree $T = (V, E)$ in which every non-terminal vertex has no more than m -children is called m -ary tree.

Full m-ary Tree: A m -ary tree $T = (V, E)$ where every internal vertex has exactly m children is called full m -ary tree.

An m -ary with $m=2$ is called binary tree

Example:

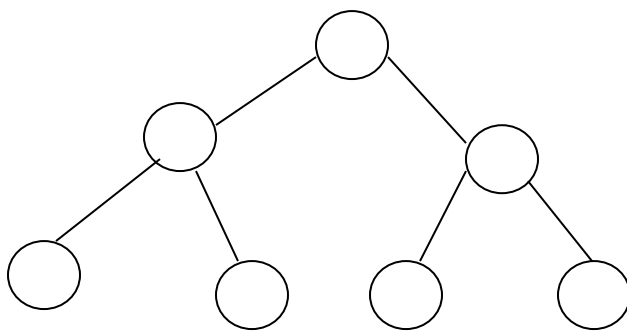


Figure : Full 2 ary tree

Theorem: A full m -ary tree with ' I ' internal vertices contain $n = mi+1$ vertices.

Proof: Let $T = (V, E)$ is a full m -ary tree with ' I ' internal vertices except the root are child of internal vertices.

Since, by definition of full m-ary tree each internal vertex has exactly m children and hence total vertices except root is equal to $m \cdot i$

Hence full binary tree T has total number of vertices $(n) = m \cdot i + 1$ including root

Theorem

A full m-ary tree with

- i. n vertices has $i = (n-1)/m$ internal vertices and $l = [(m-1) \cdot i + 1]$ leaves
- ii. i internal vertices has $n = m \cdot i + 1$ vertices and $l = (m-1) \cdot i + 1$ leaves
- iii. l leaves has $n = (l-1)/(m-1) + 1$ vertices and $i = (l-1)/(m-1)$ internal vertices

Proof: A full m-ary tree with i internal vertices has

$$n = m \cdot i + 1 \text{ vertices} \dots\dots\dots (i)$$

$$\text{and } n = i + l \quad [\text{Total vertices} = \text{no of internal vertices} + \text{leaf vertices}]$$

From eqn(i)

$$n = m \cdot i + 1 \dots\dots\dots (ii)$$

$$\text{Or } m \cdot i = n - 1$$

$$\text{Or } i = (n-1)/m \text{ proved}$$

Again putting the value of l in eqn(ii), we get

$$n = (n-1)/m + 1$$

$$\text{or } l = n - (n-1)/m$$

$$\text{or } l = (m \cdot n - n + 1)/m$$

$$\text{or } l = (n(m-1) + 1)/m$$

(ii) We know that

$$n = m \cdot i + 1$$

$$\text{and } n = i + l$$

and hence

$$m \cdot i + 1 = i + l$$

$$\text{or } l = i \cdot (m-1) + 1 \text{ Proved}$$

(ii) We know that

$$n = mi+1 \dots\dots\dots(1)$$

$$\text{and } n = l+1 \dots\dots\dots(2)$$

Solving these two equations we get

$$mi+1 = l+i$$

$$\text{or } mi-i = l-1$$

$$\text{or } i*(m-1) = l-1$$

$$i = \frac{(l-1)}{(m-1)} \text{ proved}$$

Again putting value of (i) in (1)

$$n = m \frac{(l-1)}{(m-1)} + 1$$

$$\text{or } n = \frac{(ml-1)}{(m-1)} \text{ Proved}$$

Theorem

Let $G(V,E)$ be a loop free undirected graph. Then G is a tree if there is a unique path between any two vertices of G

Proof: Let $G = (V,E)$ be a graph. Since there is a path between each pair of vertices u and v , G must be connected. Thus, to show G is a tree it remains to show that G has no cycles. Since G is loop free, it has no cycles of length 1. Suppose that G has a cycle of length greater than one, say

$$C = (v_1, v_2, v_3 \dots v_n, v_1)$$

Then any two distinct vertices of the cycle C are joined by two paths which contradict the fact that there is unique path between any two vertices of G . Hence G , has no cycles and so it is tree

Theorem

A tree with n vertices has exactly $n-1$ edges

Proof:

Let $G = (V,E)$ be a tree with n vertices. We use induction method to prove it. Let $n = 1$. i.e. has one vertex, since it has no loops, the number of edges in G is 0. This establishes that it is true for $n=1$

Let it be true for n . Now we wish to show it is true for $n+1$. Let G be a tree with $n+1$ vertices and let u be a vertex of degree n . If we remove such a vertex and the edge incident on it, then sub graph $G-u$ is still connected and has no cycles. Hence $G-u$ is a tree. However $G-u$ has n vertices, so by induction it has $n-1$ edges. Since $G-u$ has exactly one edge less than that of G , it follows that G has n edges. So summing $n+1$ vertices of G we get n edges. This completes the proof.

Theorem

In any tree G , there are at least two pendant vertices

Proof

Let $G = (V, E)$ be a tree and v_1 be a vertex of degree 1. If there is a vertex v_2 adjacent to v_1 with degree 1 then we are done. If not, we select v_3 adjacent to v_2 . Continuing in this way, we get a sequence of vertices $v_1, v_2, v_3, \dots, v_n$. Since a tree has no cycle, this sequence terminates for some k , i.e. there exists an adjacent vertex with degree 1. Thus v_1 and v_k are our desired vertices each of degree 1.